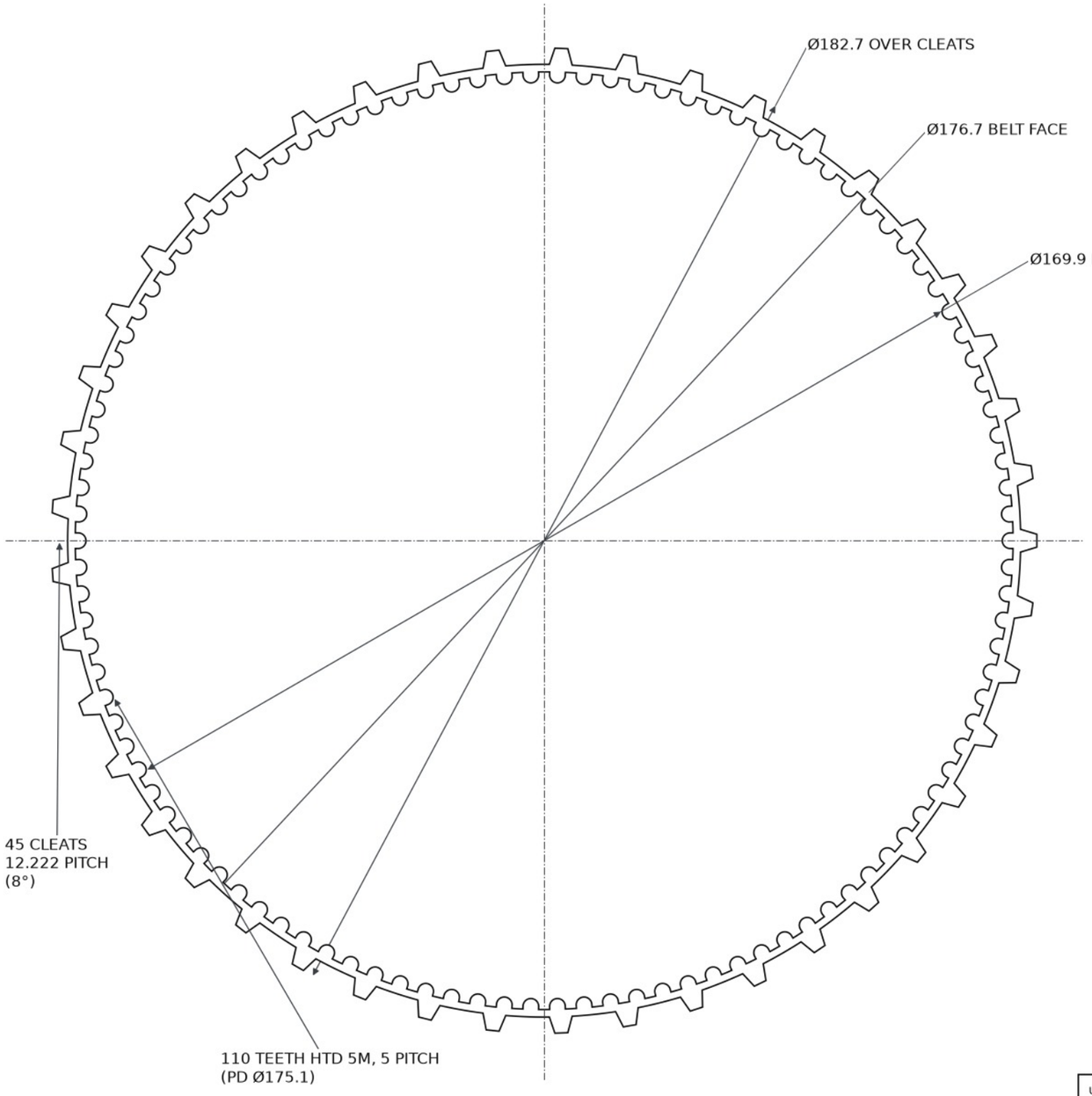
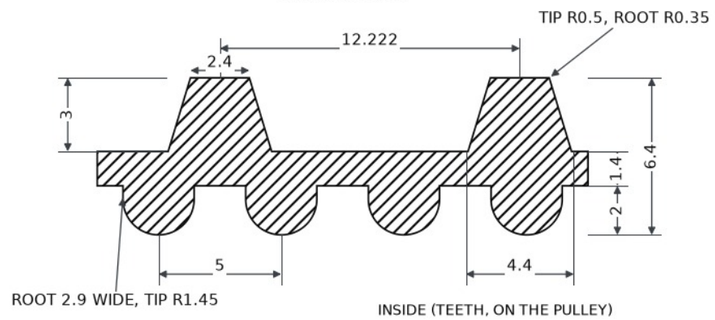
FRONT VIEW




EDGE VIEW




DETAIL: BELT SECTION, STRAIGHTENED (4:1)




NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**TRACK, 110T HTD 5M, 45 CLEATS**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

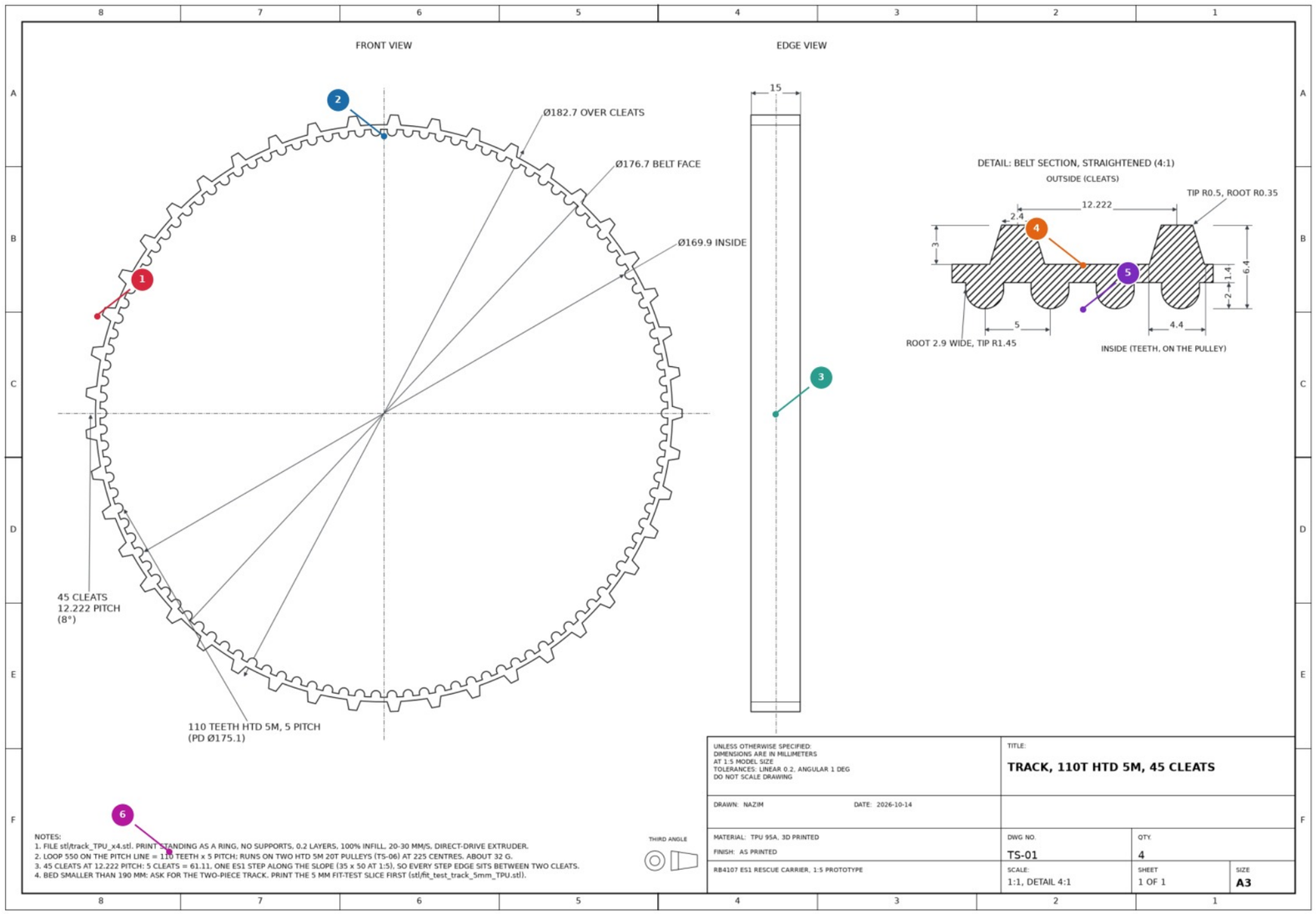
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

# How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.




**1 Cleats, outside**

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

**2 Teeth, inside**

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

**3 Edge view**

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

**4 Cleat profile (4:1)**

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

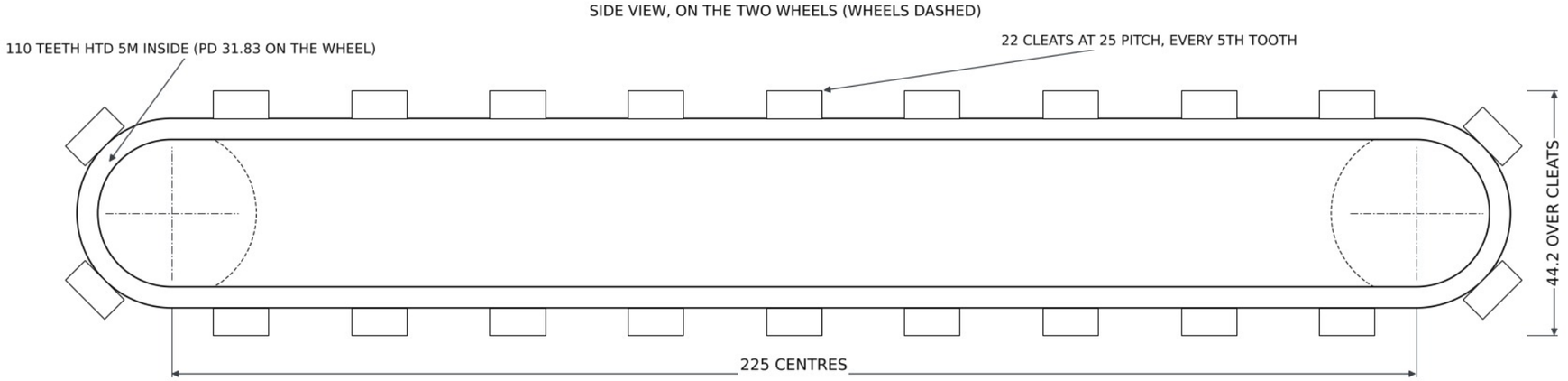
**5 Tooth profile (4:1)**

A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

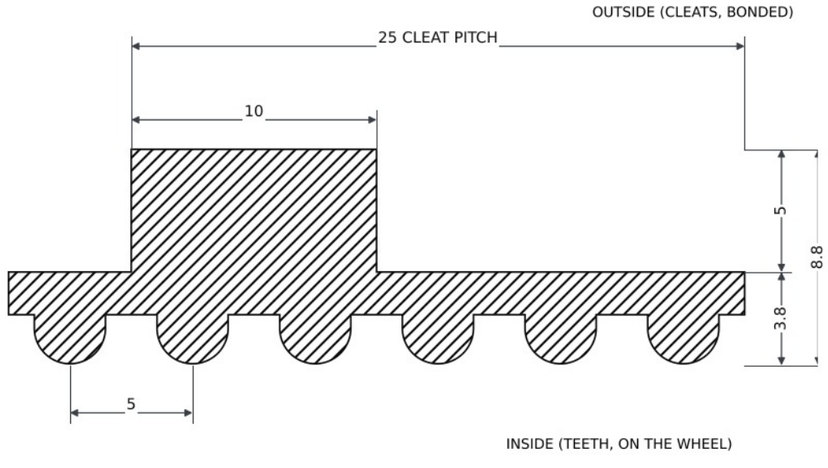
**6 Notes**

Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

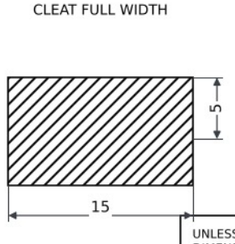
SIDE VIEW, ON THE TWO WHEELS (WHEELS DASHED)




DETAIL: SECTION ALONG THE BELT, STRAIGHTENED (4:1)




SECTION ACROSS THE BELT (2:1)




NOTES:
1. BELT: HTD 5M, 550 PITCH LENGTH (110 TEETH), 15 WIDE, E.G. SKF PHG 550-5M-15 (RS SG 058-6535). RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES.
2. CLEATS: 22 PER BELT, RUBBER 10 x 5 x 15, ONE EVERY 5TH TOOTH (25 PITCH), BONDED ON THE BELT BACK BY THE BELT SHOP (REV B BOM B10). 4 BELTS = 88 CLEATS.
3. CLEATS 5 HIGH (CUT DOWN FROM 7.9, NHIEN 19 OCT): TIPS 22.1 FROM THE AXLE, 2.9 UNDER THE DECK UNDERSIDE, SO DECK P12 NEEDS NO NOTCH.
4. ALTERNATIVE: TS-01 PRINTED TPU TRACK, 3 MM CLEATS, 5 MM UNDER THE DECK. SAME WHEELS, SAME CENTRES, SAME JOINTS.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

**TRACK BELT, BOUGHT: HTD 550-5M-15 + 22 CLEATS**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: NEOPRENE HTD BELT, RUBBER CLEATS
FINISH: AS BOUGHT, CLEATS BONDED

DWG NO. TS-01B

QTY. 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

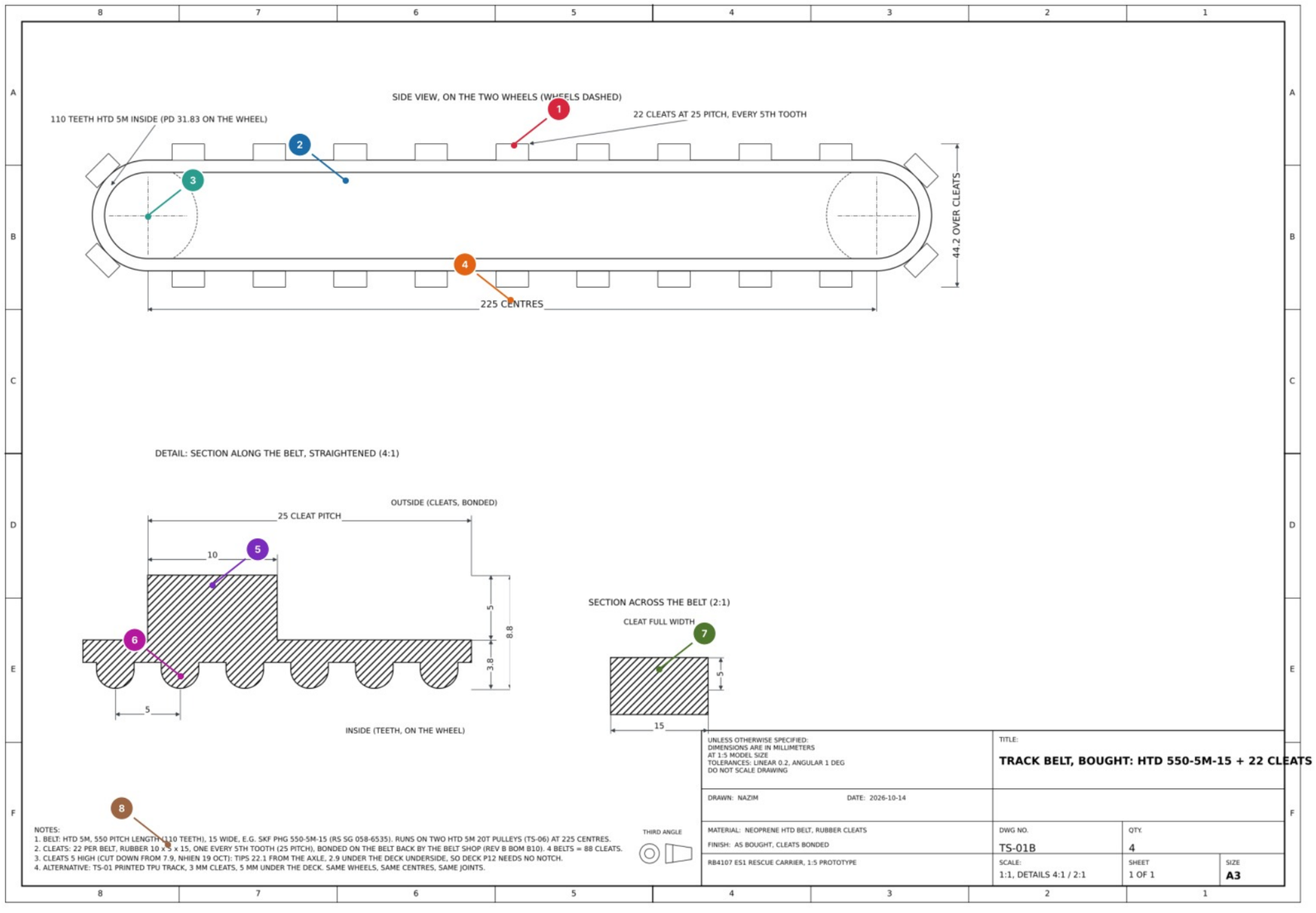
SCALE: 1:1, DETAILS 4:1 / 2:1

SHEET 1 OF 1

SIZE **A3**

# How to read TS-01B: the bought track belt (Rev B B10)

Markers below sit on a reduced copy of sheet TS-01B (previous page, full size). Numbers match the key on the right.




**1 Cleats, outside**
22 rubber blocks 10 long, 5 high (cut down from 7.9) and the full 15 width, bonded on the belt back every 25 mm (every 5th tooth). They grip the step edges. Their tips are 22.1 mm from the axle, 2.9 under the deck, so the deck needs no notch.

**2 Teeth, inside**
110 HTD 5M teeth, 5 apart, moulded into the belt. They mesh with the 20-tooth wheels, so the belt cannot slip.

**3 Hinge wheel (dashed)**
The belt wraps the driven 20T wheel on the hinge shaft at the left and the idler at the right, 225 apart: 450 straight plus the two half wraps = 550, the belt's pitch length.

**4 225 centres**
Same wheel spacing as the printed track, so the plates, slots and idlers do not change between the two versions.

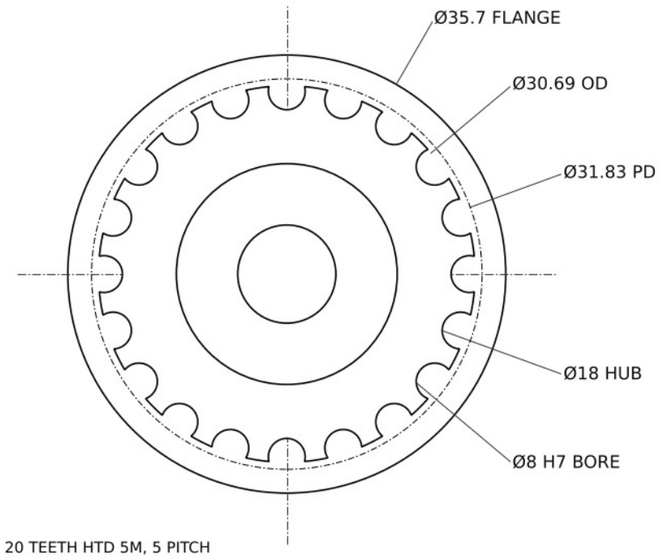
**5 Cleat profile (4:1)**
A cleat is a plain block 10 long and 5 high, bonded to the belt back; no taper. 8.8 total from tooth tip to cleat tip.

**6 Tooth profile (4:1)**
HTD 5M tooth, 2.06 deep, on a 1.74 backing: 3.8 belt thickness. This is the belt as bought.

**7 Across the belt (2:1)**
Belt and cleat are both 15 wide; the cleat spans the full width.

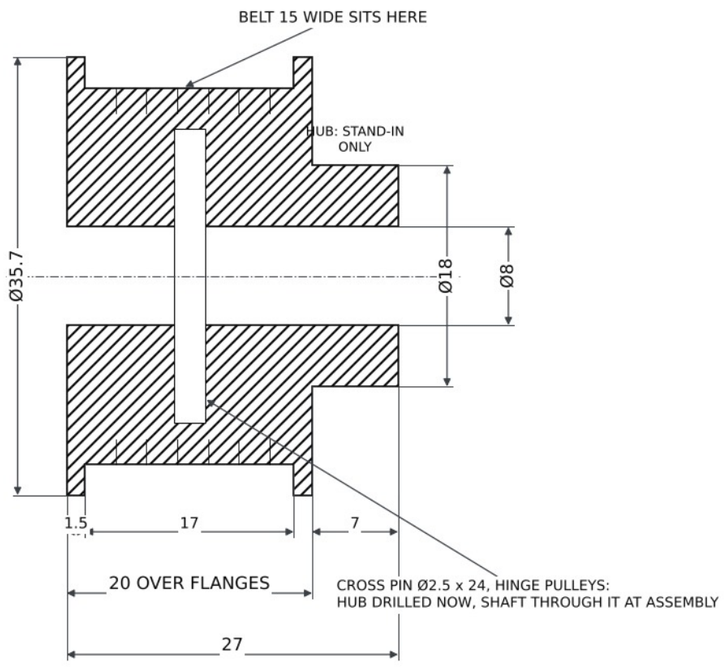
**8 Notes**
Buy SKF PHG 550-5M-15 (RS SG 058-6535) x4; a belt shop cuts and bonds 88 cleats, 5 high, so the belts fit under the deck with no notch.

FRONT VIEW




20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)




NOTES:

1. BOUGHT PART, MISUMI ECONOMY HTD 5M (H5M, NOT S5M), 20 TEETH, 15 MM BELT, FLANGES BOTH SIDES, NO HUB. HINGE WHEELS: 4 + 1 WITH BORE Ø8 TYPE H (ELSE P, IGNORE ITS SCREW). IDLERS: 4 WITH THE SMALLEST BORE, BORED Ø16 AT THE WORKSHOP. CHECK 19 OR LESS OVER THE FLANGES.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83) AND THE 15 MM BELT. ON NHIEN'S PLATES THE LANE IS 19-20 WIDE, SO NO HUB FITS: THE HUB AND SET SCREW DRAWN ARE THE PLA STAND-IN'S ONLY.
3. HINGE WHEELS: Ø2.5 CROSS HOLE AT MID-FACE (90° FROM ANY TAPPED HOLE): DRILL IT NOW, ON THE PRESS IN A V-BLOCK. AT ASSEMBLY THE SHAFT IS DRILLED THROUGH THE HUB HOLE IN PLACE. SPRING PIN ISO 8752 2.5 x 24.
4. THE PIN DRIVES AND LOCATES THE PULLEY: NO GRUB SCREW, NO E-CLIPS (TS-11). 4 x BORE Ø16 H7 THROUGH FOR 2 x 688-2RS EACH (IDLERS ON Ø8 SHOULDER SCREWS, SLEEVE 8 x 10 x 9 BETWEEN).
5. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

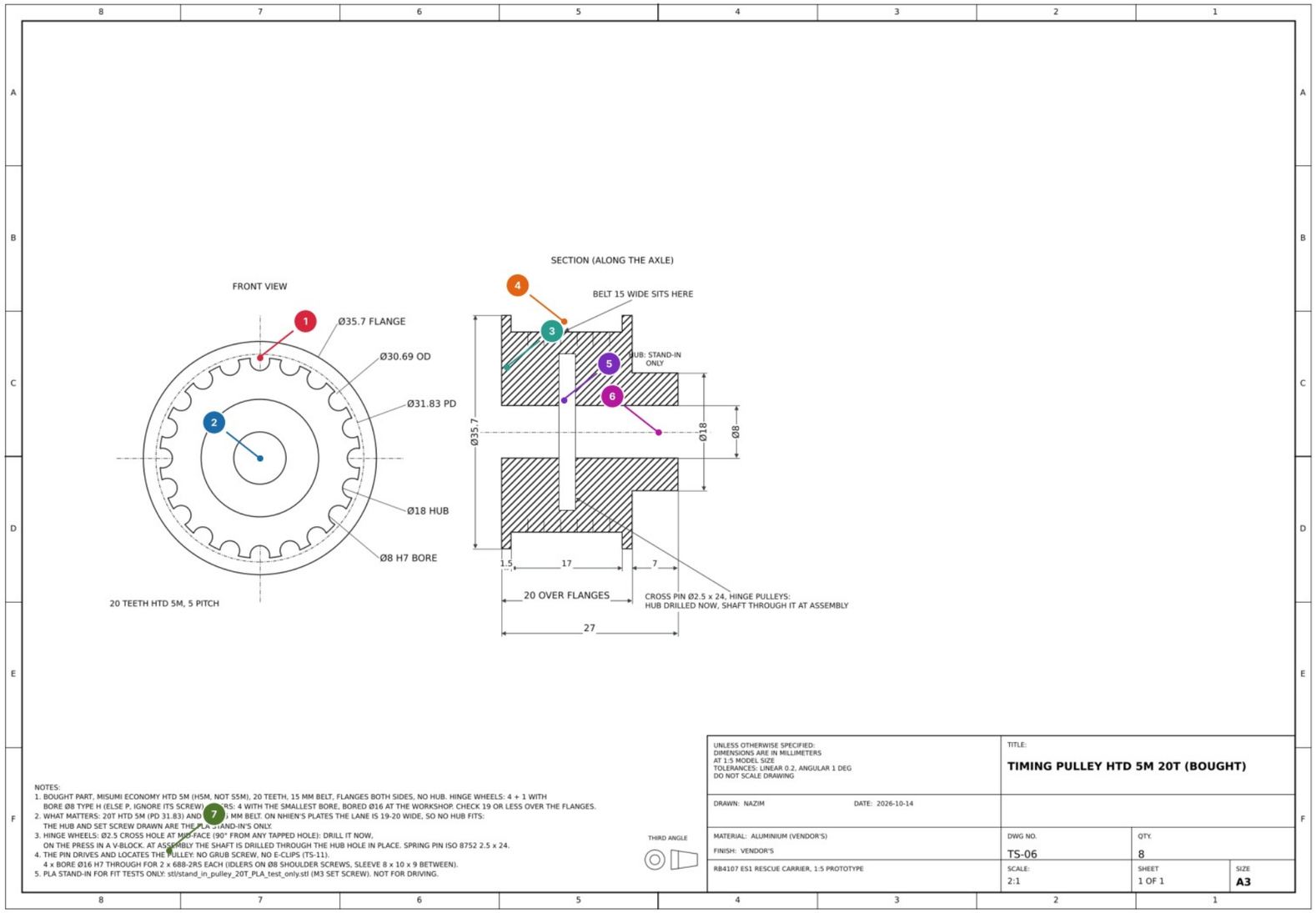
THIRD ANGLE



| | | | |
|---|---|---|---|
| UNLESS OTHERWISE SPECIFIED:<br>DIMENSIONS ARE IN MILLIMETERS<br>AT 1:5 MODEL SIZE<br>TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG<br>DO NOT SCALE DRAWING | TITLE:<br>**TIMING PULLEY HTD 5M 20T (BOUGHT)** | | |
| DRAWN: NAZIM DATE: 2026-10-14 | | | |
| MATERIAL: ALUMINIUM (VENDOR'S)<br>FINISH: VENDOR'S | DWG NO.<br>TS-06 | QTY.<br>8 | |
| RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE | SCALE:<br>2:1 | SHEET<br>1 OF 1 | SIZE<br>**A3** |

# How to read TS-06: the toothed wheel (bought, reamed, pinned)

Markers below sit on a reduced copy of sheet TS-06 (previous page, full size). Numbers match the key on the right.




**1 Teeth**
20 HTD 5M teeth on a 31.83 pitch circle: the same tooth form as the track's inside, so they mesh without slip.

**2 Bore**
Bought with a small pilot bore. The 4 hinge wheels are bought bored Ø8 to run on the plain round shaft; the 4 idlers are bored Ø16 H7 for two 688 bearings each.

**3 Flange**
Ø35.7 flanges both sides, 1.5 thick, keep the track on the wheel. 19 over the flanges, 0.5 clear of each plate in the 20 lane.

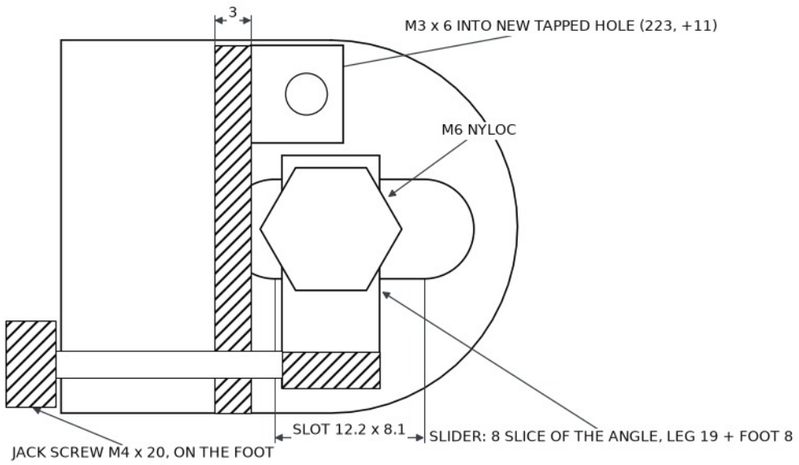
**4 Belt face**
16 between the flanges; the 15 wide track sits here.

**5 Cross pin**
Ø2.5 × 24 spring pin through wheel and shaft at mid-face: the hub is drilled on the press when the pulley is bored, the shaft is drilled through that hole at assembly, in place. It drives the wheel and holds it along the shaft: no D-flat, no grub screw, no E-clips (design review, TS-11).

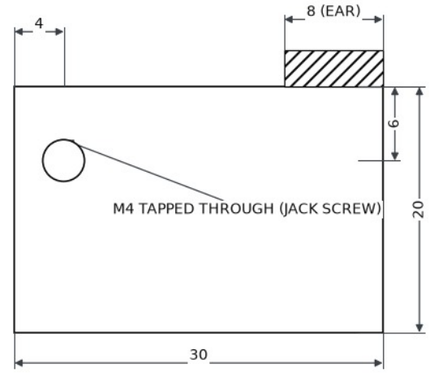
**6 Hub**
Only the printed stand-in has a hub and set screw. The bought wheel is a plain flanged disc, because the lane has no room for a hub.

**7 Notes**
Order 8 from MISUMI SG plain bore; the Professional Workshop reams and bores them; pins go in on the build day.

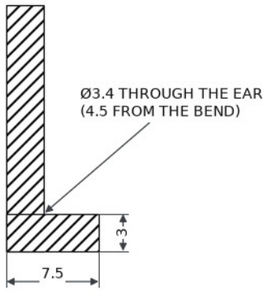
NOSE END OF P1, SEEN FROM INBOARD (THE z 55 FACE)



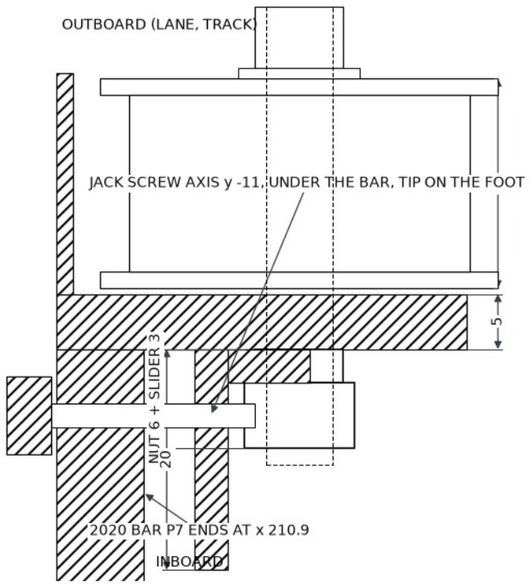

TAB: AL ANGLE 20x20x3, CUT 30, LEG A CUT TO A 7.5 x 8 EAR



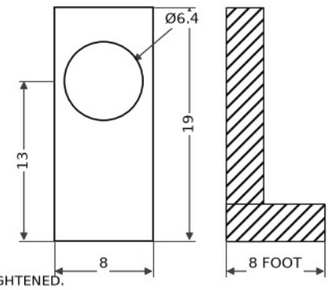

TAB, END VIEW




PLAN FROM ABOVE: PLATE, LANE, IDLER, TAB




SLIDER: 8 SLICE OF THE 20x20x3 ANGLE, FACE AND END VIEW




NOTES:
1. THE SHOULDER SCREW IS CLAMPED IN THE SLOT BY FRICTION ONLY. THE JACK SCREW SETS THE IDLER, THEN THE NUT IS TIGHTENED.
2. SLIDER REPLACES THE M6 WASHER UNDER THE NUT (3 THICK). TAB: NEW M3 TAPPED HOLE IN P1 / P2 AT x = +-223, y = +11 (ADD TO THE DXF, OR DRILL 2.5 AND TAP ON SITE).
3. FIT: IDLER INTO THE LOOP, SHOULDER SCREW THROUGH, SLIDER, NUT LOOSE, THEN THE TAB. TENSION (MACHINE ON BLOCKS): HOOK A CLEAT OF THE LOWER RUN AT MID-SPAN, PULL IT OFF ITS RAIL WITH 4 N, WIND THE JACK SCREW UNTIL IT GIVES 2-3 MM, TIGHTEN THE NUT, BACK OFF 1/4 TURN. PUSHING A RUN INWARD MOVES NOTHING: THE RAILS SIT UNDER BOTH RUNS.
4. DO NOT PUSH THE IDLER IN PAST x = 223 (RAIL END): TRAVEL 223 TO 232.6 = 9.6 MM = 19 MM OF LOOP. JACK SCREW M4 x 20 SOCKET HEAD, KEY FROM THE HINGE SIDE, UNDER THE BAR; IT PUSHES THE SLIDER'S FOOT. LEG B 1.5 OFF THE SLOT END.

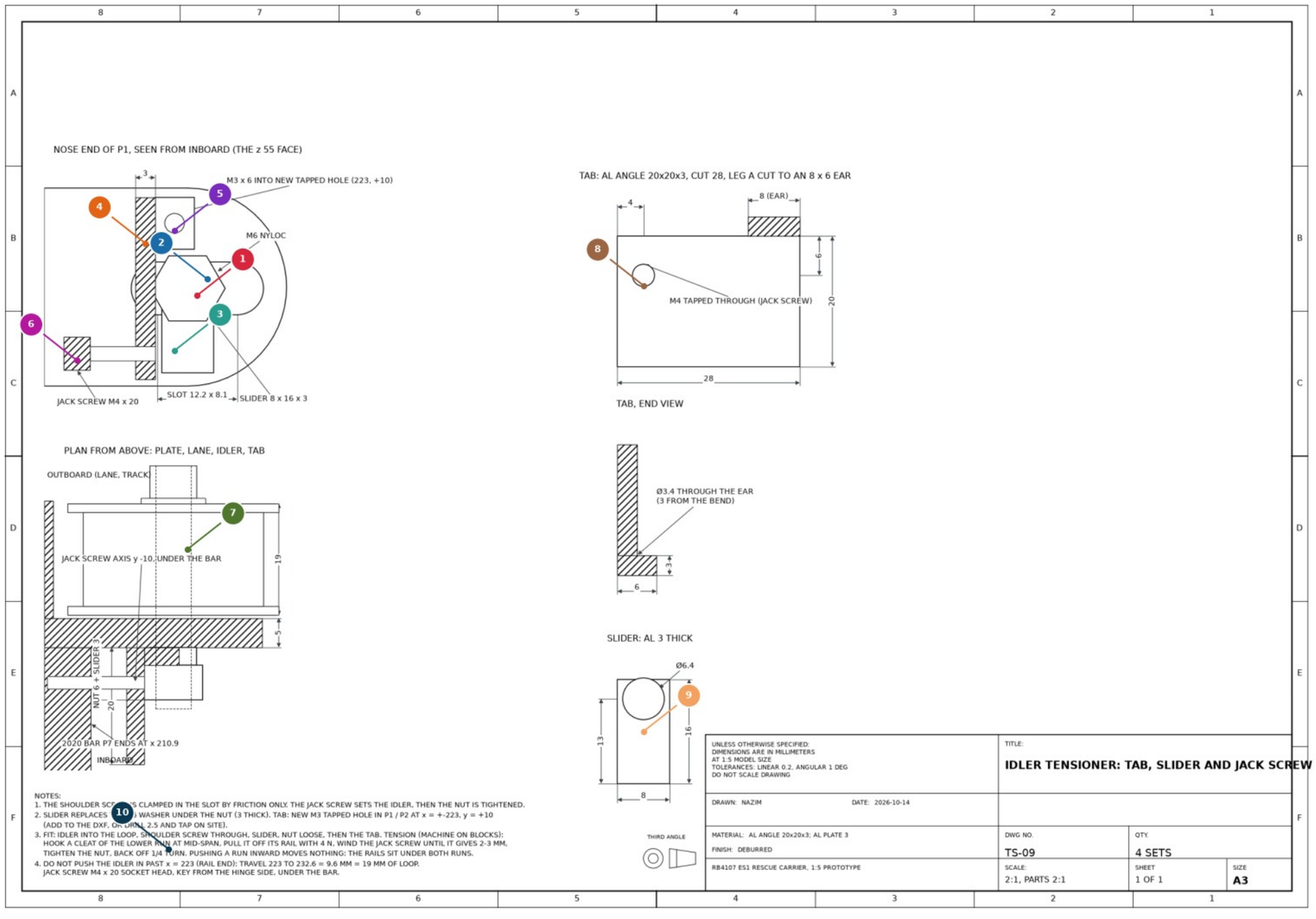
THIRD ANGLE



| | | | |
|---|---|---|---|
| UNLESS OTHERWISE SPECIFIED:<br>DIMENSIONS ARE IN MILLIMETERS<br>AT 1:5 MODEL SIZE<br>TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG<br>DO NOT SCALE DRAWING | TITLE:<br>**IDLER TENSIONER: TAB, SLIDER AND JACK SCREW** | | |
| DRAWN: NAZIM DATE: 2026-10-14 | | | |
| MATERIAL: AL ANGLE 20x20x3; AL PLATE 3<br>FINISH: DEBURRED | DWG NO.<br>TS-09 | QTY.<br>4 SETS | |
| RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE | SCALE:<br>2:1, PARTS 2:1 | SHEET<br>1 OF 1 | SIZE<br>**A3** |

# How to read TS-09: the idler tensioner

Markers below sit on a reduced copy of sheet TS-09 (previous page, full size). Numbers match the key on the right.




### 1 Slot and shoulder screw

The idler's Ø8 shoulder screw sits in the plate's 12.2 x 8.1 slot. Sliding it outward tightens the track; before this sheet only the nut's friction held it.

### 2 Nyloc nut

M6 nyloc on the inboard side. Tightened last, after the jack screw has set the tension.

### 3 Slider

An 8 mm slice of the same 20 x 20 x 3 angle in place of the M6 washer: a 19 mm leg on the plate with the Ø6.4 hole, and an 8 mm foot standing off the plate. The jack screw pushes on the foot, so nothing pushes on the nut.

### 4 Tab, leg B

The angle's long leg stands off the plate, inboard. The M4 jack screw threads through it along the slot direction.

### 5 Ear and M3

The short leg is cut down to a 7.5 x 8 ear above the slot and bolted to the plate with one M3 x 6 into a new tapped hole at (223, +11). Leg B stands 1.5 mm clear of the slot end, so the nut's corners never touch it.

### 6 Jack screw

M4 x 20 socket head. Wind it in until the lower run, hooked at a cleat at mid-span and pulled off its rail with 4 N (machine on blocks), gives 2 to 3 mm; pushing a run inward moves nothing because the rails sit under both runs, tighten the nut, then back it off a quarter turn.

### 7 Plan: the lane

Seen from above: plate P1 at z 55 to 60, the idler wheel in the lane with the screw head outside, the nut, slider and tab inboard, the 2020 bar ending at x 210.9 just behind the tab.

### 8 Tab, flat

Angle 20 x 20 x 3 cut 30 long; the M4 hole is 4 from the bottom end and 6 from the bend; the ear is the top 8 of the other leg, trimmed to 7.5 wide.

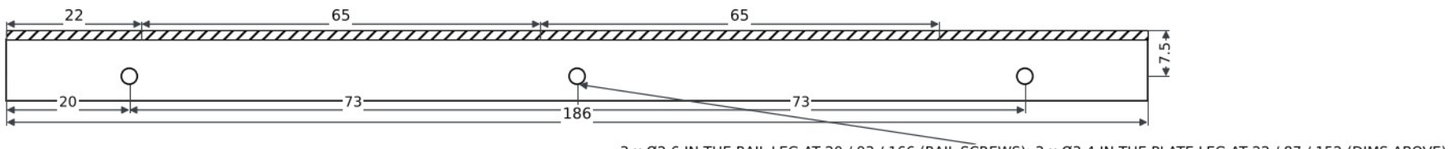
### 9 Slider

8 mm slice of the angle: leg 19 with the Ø6.4 hole 13 from the bottom, foot 8 bent inboard.

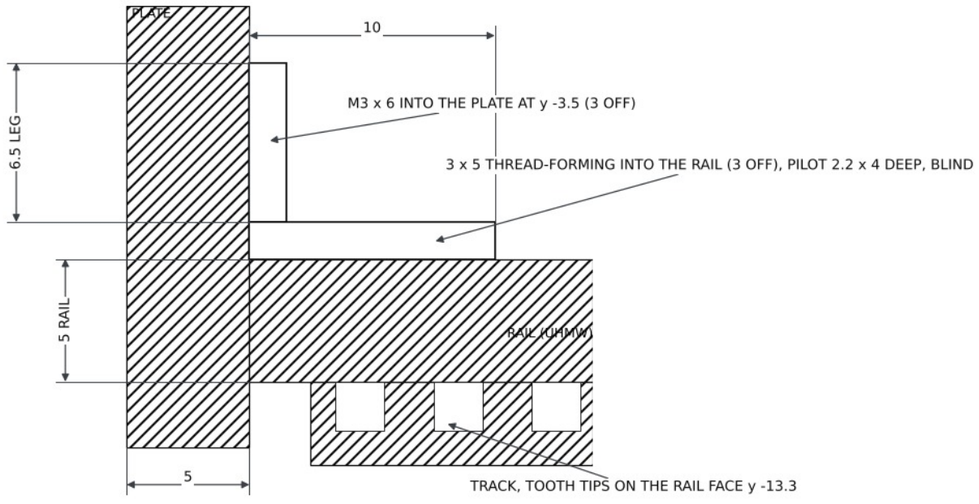
### 10 Notes

Four sets: nose and rear idler on each side. The rear tab mirrors the front; the jack screw always pushes outward, never in past x 223.

PLAN OF ONE ANGLE, 1:1 (RAIL LEG FLAT)




SECTION ACROSS THE LANE AT THE LOWER RAIL (4:1): PLATE, ANGLE, RAIL, TRACK




NOTES:

1. WHY: EACH UHMW RAIL HUNG ON 4 SMALL CLIPS (10 x 14.6 FEET). THE ANGLE RUNS THE FULL RAIL LENGTH: 186 x 10 OF CONTACT ON THE PLATE AND ON THE RAIL INSTEAD OF 4 SMALL PATCHES.
2. TRIM EACH RAIL FROM 189 TO 186 (3 OFF THE IDLER END): THE IDLER PULLEY FLANGE REACHES x 207 AT ITS INNERMOST SLOT POSITION AND THE RAIL ENDED AT 207 (TS-11).
3. PLATE LEG CUT TO 6.5 SO THE UPPER AND LOWER ANGLES DO NOT MEET ON THE PLATE'S CENTRELINE. 3 x Ø3.4 IN THE PLATE LEG AT x 40, 105, 170 FROM THE HINGE, y +-3.5: SIX NEW M3 HOLES IN P1 / P2 (THE OLD CLIP HOLES ON THE CENTRELINE ARE NOT USED). 3 x Ø2.6 IN THE RAIL LEG AT 20, 93, 166 FROM THE RAIL'S HINGE END (x 38, 111, 184), 7.5 FROM THE PLATE FACE: 3 x 5 THREAD-FORMING SCREWS. DO NOT DRILL THE OLD CLIP PILOTS IN THE RAIL: THEY OVERLAP THE NEW ONES AT x 38 / 40 AND 184 / 185.
4. THE ANGLE SITS ON THE RAIL FACE AWAY FROM THE TRACK: UNDER THE UPPER RAIL, ON TOP OF THE LOWER RAIL. THE TRACK'S TOOTH TIPS RIDE ON THE OTHER FACE (y 13.3).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

**RAIL CARRIER ANGLE (REPLACES RS 427-991 CLIPS)**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: AL ANGLE 10x10x1.5

FINISH: DEBURRED

DWG NO. TS-10

QTY. 8

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

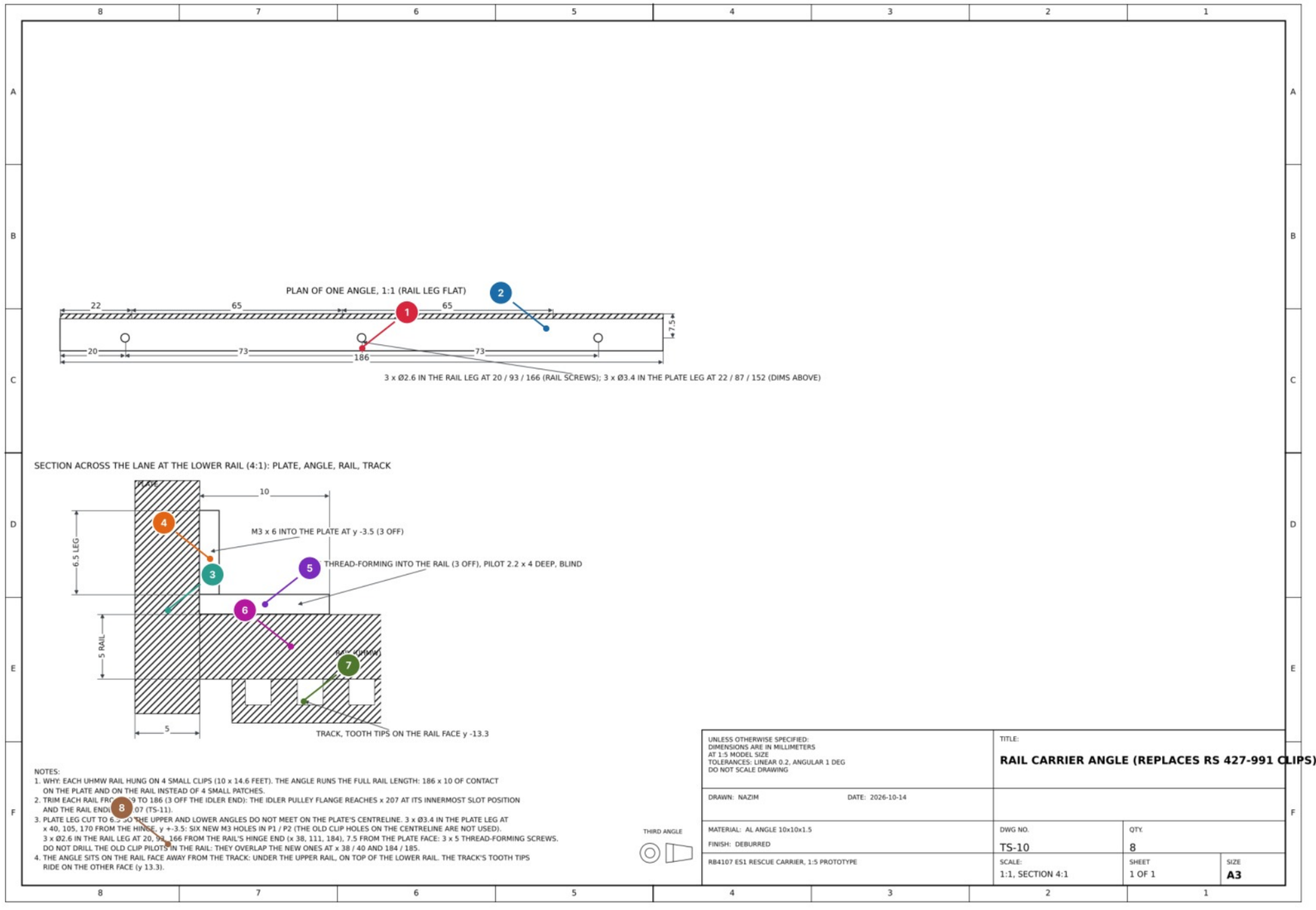
SCALE: 1:1, SECTION 4:1

SHEET 1 OF 1

SIZE **A3**

# How to read TS-10: the rail carrier angle

Markers below sit on a reduced copy of sheet TS-10 (previous page, full size). Numbers match the key on the right.




**1 Angle, plan**
One 10 × 10 × 1.5 aluminium angle per rail, 186 long: three Ø2.6 holes in the rail leg for the thread-forming screws at 20, 93 and 166 from the rail's hinge end, 7.5 from the plate face, three Ø3.4 in the plate leg at 22, 87 and 152 from the hinge end (x 40, 105, 170), 3.5 from the leg's edge, for M3 x 6 into six new tapped holes in the plate.

**2 Plate leg**
Seen edge on above the rail leg. It is cut down to 8 so it stays inside the 15.4 plate height.

**3 Plate**
P1 or P2, 5 thick. The three M3 × 6 go through the angle into the plate's tapped holes, the same ones the four RS clips used.

**4 Angle on the plate**
The 8 leg lies flat on the plate's lane-side face, full length: 186 × 8 of contact instead of four 10 × 14.6 clip feet.

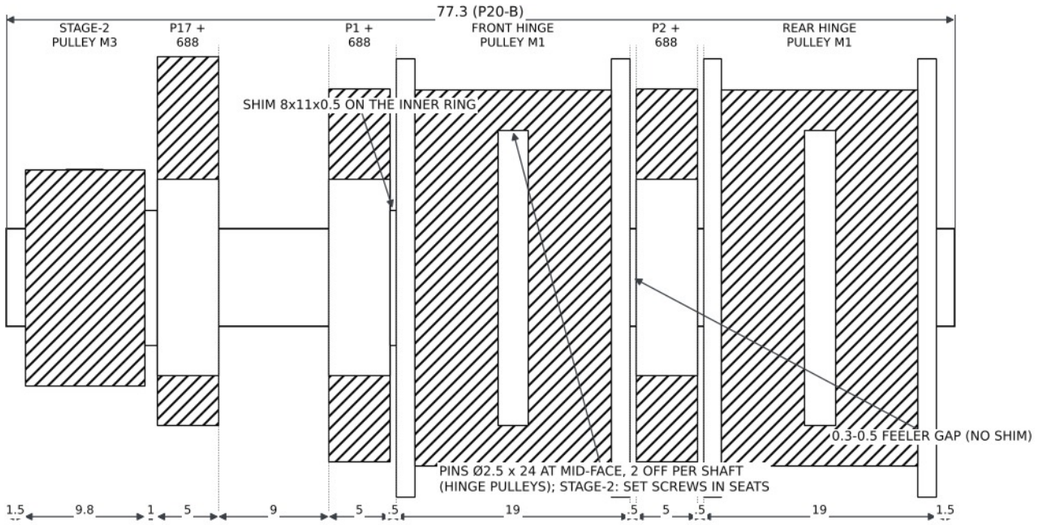
**5 Angle on the rail**
The 10 leg lies on the rail's face away from the track: on top of the lower rail, under the upper one. Three 3 × 5 thread-forming screws into blind Ø2.2 pilots 4 deep: the rail is only 5 thick and its far face carries the track, so a longer screw would come out of the sliding face.

**6 Rail**
UHMW strip 20 wide, 5 thick, trimmed from 189 to 186 at the idler end so the idler wheel's flange can never reach it.

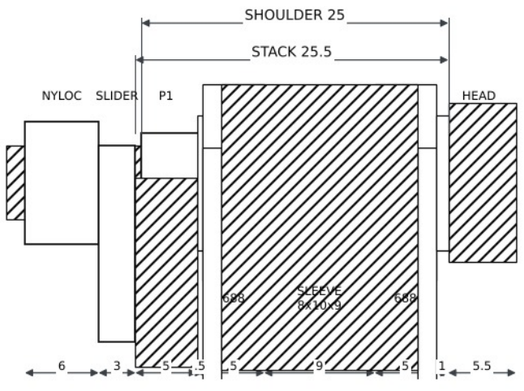
**7 Track**
The track's lower run rides tooth-tips-down on the rail's other face at y -13.3; the angle never touches the track.

**8 Notes**
Eight angles from 1.6 m of stock; the RS 427-991 clips and their 24 M3 × 6 are no longer needed (keep the thread-forming screws).

SECTION ALONG THE HINGE SHAFT P20-B, FROM THE CENTRE (z 28.7) OUTWARD TO 106. PINS REPLACE THE E-CLIPS.




SECTION ALONG THE NOSE IDLER SCREW (ISO 7379 8 x 25 M6), INBOARD TO OUTBOARD




NOTES:
1. FITS: 688-2RS SEATS IN P1, P2, P17 LASER-CUT Ø15.8 THEN REAMED Ø16 H7, RETAINED WITH LOCTITE 641. HINGE PULLEYS Ø8 H7 ON A PLAIN Ø8 h7 ROUND SHAFT (NO D-FLAT, NO GROOVES).
2. IDLER PULLEYS Ø16 H7, 688-2RS LIGHT PRESS. SLOTS 8.1 (+0.1 / 0) FOR THE Ø8 f9 SHOULDER. LANE 20.0 +-0.2 = (160 - 110) / 2 - 5 FROM THE BAR CUT LENGTHS: CUT BARS TO +-0.1 AND SQUARE.
3. AXIAL LOCATION ON THE HINGE SHAFT: PINS, NOT E-CLIPS. PIN THE FRONT HINGE PULLEY FIRST: ONE 8x11x0.5 SHIM AGAINST THE P1 BEARING, 0.3-0.5 FEELER GAP TO THE P2 BEARING. THAT FIXES THE SHAFT.
4. THEN PIN THE REAR PULLEY (0.3-0.5 GAP TO ITS BEARING). STAGE-2 PULLEY (1 MM SHIM AGAINST THE P17 BEARING): HUB TYPE, 2 CONE-POINT SET SCREWS IN SEATS 1 MM DEEP, LOCTITE 243. ITS HUB (NOT DRAWN) RUNS ON THE SHAFT: SHAFT LENGTH GROWS BY THE HUB LENGTH, TBC. SHIMS BEAR ON INNER RINGS ONLY.
5. IDLER SCREW: SHOULDER 25 vs STACK 25.5 (HEAD WASHER 1 + BEARINGS 10 + SLEEVE 9 + SHIM 0.5 + PLATE 5): THE NUT CLAMPS THE INNER-RING STACK AGAINST THE PLATE, AS INTENDED. KEEP THE 0.5 SHIM.
6. IDLER FLANGE MEETS THE RAIL END IF THE IDLER IS PUSHED IN PAST x 223: RAILS TRIMMED TO 186 (TS-10), TENSIONER TS-09 ONLY EVER PUSHES OUTWARD.

THIRD ANGLE

## FITS AND CLEARANCES

| | | |
|---|---|---|
| P1 / P2 / P17 bearing seat | laser Ø15.8, ream Ø16 H7 | 688-2RS, Loctite 641 |
| hinge pulley M1, stage-2 M3 bores | Ø8, bought bored | M1 pinned, M3 set screws in seats |
| idler pulley bore M2 | bore Ø16 H7 | 2 x 688-2RS, light press |
| plate slot | 8.1 +0.1 / 0 wide | shoulder Ø8 f9 |
| lane, plate face to plate face | 20.0 +-0.2 | pulley 19.0: 0.5 each side |
| pulley to bearing gaps | 0.5 shim + 0.3-0.5 feeler | never on the outer ring |
| rail end to idler flange | rail 186, idler >= x 223 | 2 mm minimum |
| deck over the front track | 25 above axis | printed 19.7 / belt 25.0 |

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
**STACK-UPS AND FITS: HINGE SHAFT AND IDLER SCREW**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE PARTS
FINISH: -

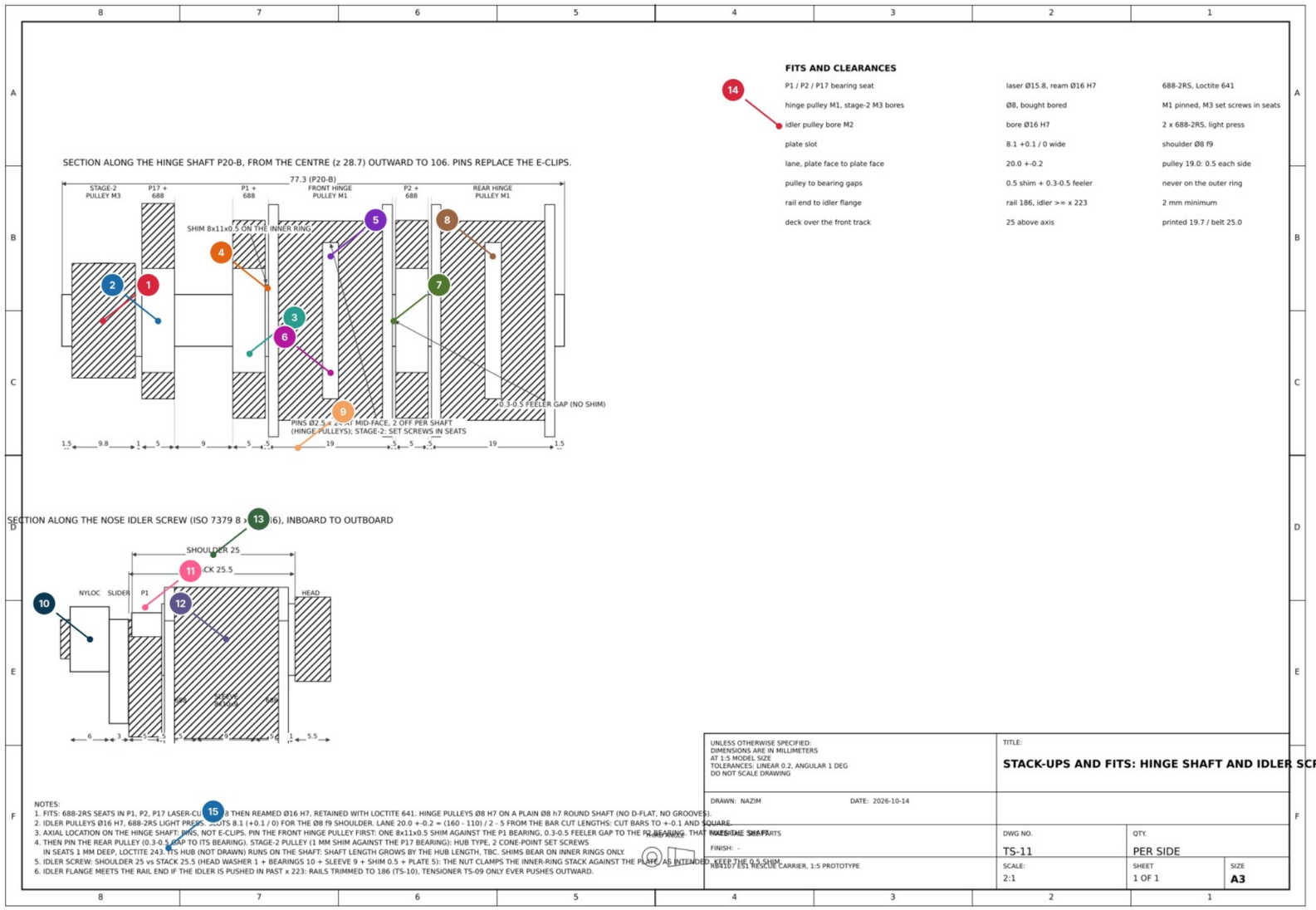
DWG NO. TS-11
QTY. PER SIDE

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 2:1
SHEET 1 OF 1
SIZE **A3**

# How to read TS-11: the spacing between parts

Markers below sit on a reduced copy of sheet TS-11 (previous page, full size). Numbers match the key on the right.




**1 Stage-2 pulley**
The HTD 5M drive pulley on the hinge shaft, z 30.2 to 40, held by two cone-point set screws in seats drilled 1 mm into the shaft, with a 1 mm shim between it and the P17 bearing. Its hub (not drawn) makes the shaft longer, by an amount still to confirm.

**2 P17 and its bearing**
The hinge-shaft support on the middle bar, bearing at z 41 to 46. 9 mm of bare shaft follow before the front plate.

**3 P1 and its bearing**
Front plate at z 55 to 60; the 688 sits flush in a reamed Ø16 H7 seat with Loctite 641.

**4 Shim 0.5**
An 8 × 11 × 0.5 shim on the bearing's inner ring, so the pulley face never touches the outer ring. The pulley is pinned hard against it.

**5 Front hinge pulley**
19 over the flanges in the 20 lane, z 60.5 to 79.5. Pinned to the shaft first: it is what locates the whole shaft between the two plate bearings.

**6 Cross pin**
Ø2.5 × 24 spring pin at mid-face, drilled through pulley and shaft together. Two per shaft, in the front and rear hinge pulleys; the stage-2 pulley uses set screws instead.

**7 Feeler gap**
0.3 to 0.5 mm between the front pulley and the P2 bearing, set with a feeler gauge before pinning. This is the shaft's total end float.

**8 Rear hinge pulley**
Outboard of P2 at z 85.5 to 104.5, pinned with its own 0.3 to 0.5 gap to the bearing. The shaft ends 1.5 past it; no E-clip.

**9 Station chain**
Every width along the shaft from the centre outward. Sum from 28.7: the 77.3 mm half shaft P20-B.

**10 Idler: nut**
M6 nyloc inboard, 6 tall, then the 3 mm slider of TS-09.

**11 Idler: plate**
P1, 5 thick; the shoulder's end sits 0.5 inside it, so the M6 washer face never touches the slot.

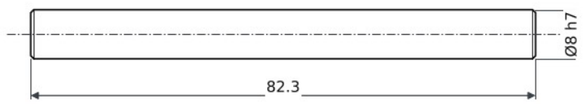
**12 Idler: bearings and sleeve**
Two 688s with the 8 × 10 × 9 sleeve between their inner rings, inside the Ø16 H7 bore of the idler wheel.

**13 Shoulder vs stack**
Shoulder 25 against a 25.5 stack: the nut clamps washer, inner rings, sleeve, shim and plate together. That is correct; do not leave the shim out.

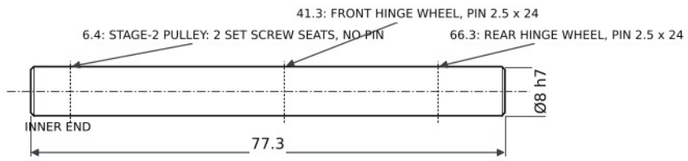
**14 Fits table**
The fit or clearance for every mating pair that matters, and what the laser, reamer or saw has to hold.

**15 Notes**
The order of pinning and the shim rule in words.

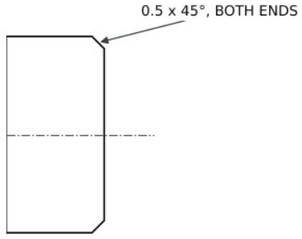
SHAFT P20-A, 82.3 + HUB (TBC), QTY 1 (1:1)



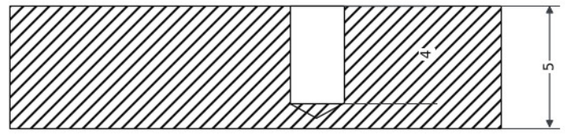

SHAFT P20-B, 77.3 + HUB (TBC), QTY 1 (1:1). POSITIONS NOMINAL, DRILLED AT ASSEMBLY



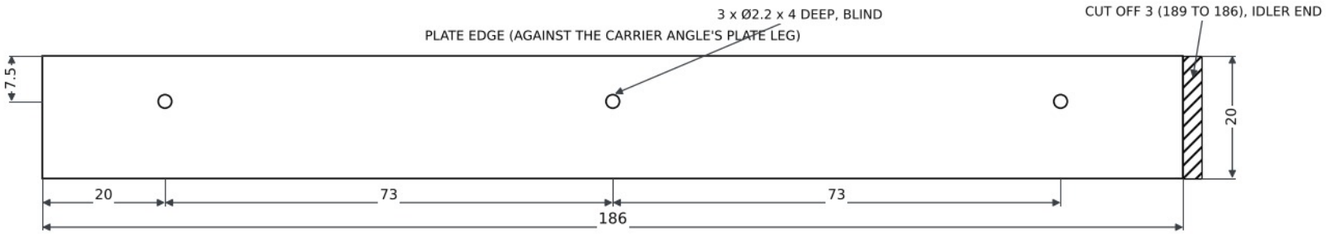

DETAIL: SHAFT END (4:1)




SECTION THROUGH A PILOT (4:1)




RAIL P3-P6, DRILLING FACE UP (1:1). PLATE EDGE AT THE TOP, HINGE END ON THE LEFT




NOTES:
1. SHAFTS: CUT BOTH FROM ONE 300 MM Ø8 h7 GROUND SHAFT: A = 82.3, B = 77.3 (+-0.1), EACH PLUS THE STAGE-2 PULLEY'S HUB LENGTH (TBC 20 OCT):
   DO NOT CUT UNTIL THE HUB IS CONFIRMED. FACE SQUARE, CHAMFER 0.5 x 45° BOTH ENDS. PLAIN ROUND: NO D-FLAT, NO E-CLIP GROOVES. KEEP THE GROUND SURFACE CLEAN: IT RUNS IN THE 688 BEARINGS.
2. DO NOT DRILL THE Ø2.5 PIN HOLES NOW. AT ASSEMBLY EACH ONE IS DRILLED THROUGH THE SHAFT USING THE PULLEY'S HUB HOLE AS THE GUIDE,
   AFTER THE SHIM AND FEELER GAP ARE SET (TS-11). THE PIN POSITIONS SHOWN ON B ARE NOMINAL, FOR REFERENCE ONLY; A IS THE SAME STACK.
3. RAILS: MISUMI UPA-189-20-5. CUT 3 OFF THE IDLER END: 186 LONG. THE SAME PART SERVES AS UPPER AND LOWER RAIL: DRILL THE FACE
   AWAY FROM THE TRACK, WHERE THE CARRIER ANGLE SITS (TS-10). MARK THE HINGE END AND THE PLATE EDGE ON EACH RAIL BEFORE DRILLING.
4. PILOTS Ø2.2 x 4 DEEP, BLIND, WITH A DEPTH STOP: THE OTHER FACE IS WHAT THE TRACK SLIDES ON AND MUST STAY UNBROKEN.
   DO NOT DRILL THE REV B SHEET 18 PILOTS: THEY OVERLAP THESE AT 20 / 22 AND 166 / 167.

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE: **HINGE SHAFTS A + B, RAIL TRIM AND PILOTS**

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: Ø8 h7 GROUND STEEL; UHMW-PE 20 x 5

FINISH: DEBURRED

DWG NO. TS-12

QTY. 2 SHAFTS, 8 RAILS

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAILS 4:1

SHEET 1 OF 1

SIZE **A3**